1. SOA Design Principles (L R A A D C S S)

L – Loose Coupling (Design services with minimal dependencies so changes in one service do not impact others.)

R – Reusability (Build services that can be reused across multiple applications to avoid duplication.)

A – Autonomy (Each service should operate independently and manage its own logic and data.)

A – Abstraction (Hide internal implementation and expose only what is necessary through APIs.)

D – Discoverability (Ensure services are easy to find using documentation or registries.)

C – Composability (Combine services together to create larger business workflows.)

S – Statelessness (Each request should be independent without relying on stored session state.)

S – Scalability (Design services to scale independently based on demand.)

2. AWS CAF (B P G P S O)

B – Business (Align cloud adoption with business goals and ensure it delivers value.)

P – People (Develop the skills, roles, and culture required for cloud adoption.)

G – Governance (Apply policies and controls to manage risks and ensure compliance.)

P – Platform (Provide the cloud infrastructure, applications, and data environment.)

S – Security (Protect systems and data using identity, encryption, and monitoring.)

O – Operations (Run and manage systems effectively through monitoring and incident response.)

3. AWS Well-Architected Pillars (O S R P C S)

O – Operational Excellence (Run systems efficiently using automation, monitoring, and continuous improvement.)

S – Security (Protect data and systems through strong identity and threat detection controls.)

R – Reliability (Ensure systems recover quickly from failures and remain available.)

P – Performance Efficiency (Use resources efficiently and scale to meet demand.)

C – Cost Optimization (Minimize cost by using only the resources you need.)

S – Sustainability (Reduce environmental impact through efficient resource usage.)

4. AWS Cloud Design Principles (S T A M M D C U)

S – Stop guessing capacity (Use metrics and Auto Scaling instead of predicting capacity needs.)

T – Test in production (Validate systems using real-world techniques like canary releases.)

A – Automate everything (Automate infrastructure, deployments, and operations.)

M – Make small, frequent changes (Reduce risk by releasing small and frequent updates.)

M – Monitor everything (Continuously track logs, metrics, and system behavior.)

D – Design for failure (Build systems assuming failures will happen and ensure resilience.)

C – Cost optimization (Continuously manage and reduce cloud costs.)

U – Use serverless when possible (Leverage managed services to reduce operational effort.)

AWS Fargate

👉 “Run containers without managing servers”

Serverless compute for containers (ECS/EKS). No EC2 management needed. You define CPU/memory and AWS runs and scales it.

AWS Cost Explorer

👉 “Track and analyze AWS costs”

Shows where your AWS money is going and how discounts are applied.

AWS Glue Studio

👉 “Visual ETL builder”

Design, transform, and prepare data workflows with minimal code.

AWS Backup

👉 “Centralized backup management”

Automates backups and restores across AWS services.

AWS DataSync

👉 “Automated data transfer”

Moves data between on-prem, AWS, or AWS storage services (fast and repeatable).

AWS Application Migration Service

👉 “Lift-and-shift servers”

Replicates and migrates on-prem servers to EC2.

AWS Systems Manager

👉 “Manage servers centrally”

Control EC2 and on-prem servers without SSH/RDP. Useful for hybrid environments.

Amazon ECR

👉 “Container image storage”

Stores and manages Docker images.

Amazon ECS

👉 “Container orchestration”

Runs and manages containers at scale.

Amazon EMR

👉 “Big data processing”

Processes large datasets using Hadoop, Spark, etc.

Amazon Lex

👉 “Build chatbots”

Creates conversational interfaces using voice and text.

ELB Health Check

👉 “Traffic only to healthy instances”

Periodically checks instances and routes traffic only to healthy ones.

Compute Savings Plans

👉 “Usage-based discount model”

Lower costs in exchange for committing to usage over time.

AWS Transfer Family

👉 “Managed file transfer”

Supports SFTP/FTP/FTPS into S3 or EFS.

Amazon Chime

👉 “Meetings and chat”

Video calls, messaging, collaboration tool.

Amazon Connect

👉 “Cloud contact center”

Build customer support systems with call routing.

Amazon OpenSearch

👉 “Search and analytics”

Search, analyze, and visualize large datasets (logs, search engines).

Amazon Pinpoint

👉 “User engagement”

Send targeted messages (email, SMS, push notifications).

AWS CodeDeploy

👉 “Automated deployments”

Deploy applications with rolling or blue/green strategies.

AWS Security Hub

👉 “Security alerts dashboard”

Centralizes findings from multiple AWS security services.

Amazon Transcribe

👉 “Speech-to-text”

Converts audio into written text.

Amazon Polly

👉 “Text-to-speech”

Converts text into natural-sounding speech.

Amazon Rekognition

👉 “Image/video analysis”

Detects objects, faces, and text in images/videos.

Amazon SNS (Simple Notification Service)

👉 “Pub/sub messaging”

Sends notifications to multiple subscribers (email, SMS, etc.).

Amazon SES (Simple Email Service)

👉 “Email service”

Send and receive emails at scale.

Amazon WorkMail

👉 “Business email service”

Managed email and calendar system.

Concierge Support

👉 - Billing support (invoices, payments, charges)

- Account support (account setup, upgrades, limits)
- General AWS guidance (which services to use, basic help)
- NOT deep technical troubleshooting

Available in:

→ Business Support Plan

→ Enterprise Support Plan

AWS Professional Services

👉 “Expert consulting”

AWS experts help design and implement solutions on aws.

AWS Transit Gateway

👉 “Central network hub”

Connects multiple VPCs and on-prem networks (hub-and-spoke architecture).

Virtual Private Gateway (VGW)

👉 “VPN endpoint in AWS”

Connects VPC to on-prem via Site-to-Site VPN.

Amazon Route 53

👉 “DNS resolver”

Routes traffic and resolves domain names, including hybrid setups.

AWS Site-to-Site VPN

👉 “Encrypted tunnel”

Secure connection between on-prem network and AWS over internet.

AWS Direct Connect

👉 “Private connection”

Dedicated network link to AWS (not over internet).

Amazon Cognito Identity Pool

👉 “Temporary credentials”

Provides temporary AWS credentials to users (even unauthenticated).

AWS KMS

👉 “Managed encryption keys”

AWS-managed key service with user-defined permissions.

AWS CloudHSM

👉 “Dedicated key hardware”

Single-tenant hardware for full control of encryption keys.

Service and Communications Protection

👉 “Customer secures network config”

You handle Security Groups, NACLs, firewall rules.

AWS CodeBuild

👉 “Build and test code”

Compiles code, runs tests, produces artifacts (CI).

Amazon EFS

👉 “Shared Linux file storage”

Scalable storage accessed by multiple instances simultaneously.

Amazon FSx

👉 “Managed file systems”

Provides Windows/Linux file systems (e.g., central Windows file server).

S3 Standard

👉 “Hot storage”

High availability, no retrieval fees.

S3 One Zone-IA

👉 “Cheap low-latency storage”

Stored in one AZ, lower cost but less resilient.

S3 Intelligent-Tiering

👉 “Automatic tiering”

Moves data between storage tiers automatically, no retrieval fee.

Amazon Athena

👉 “Query S3 with SQL”

Serverless querying directly on S3 data.

Amazon Redshift Spectrum

👉 “Query S3 via Redshift”

Extends Redshift queries to S3 data.

AWS Trusted Advisor

👉 “Best practice checker”

Evaluates environment across:

- Cost Optimization
- Performance
- Security
- Fault Tolerance
- Service Limits

AWS CAF – Launch Phase

👉 “Start delivering value”

Deploy workloads and gather feedback.

AWS CAF – Security Perspective

👉 “Secure cloud adoption”

Focuses on identity, data protection, and risk management.

Performance Efficiency Pillar

👉 “Right resources, right time”

Optimize performance using scaling and modern services.

Operational Excellence Pillar

👉 “Run systems well”

Use automation, game days, and incremental deployments.

Transfer Family + EFS

👉 “File ingestion + shared storage”

Transfer Family receives files, EFS provides shared access.

Enterprise Support

👉 “Highest support tier”

Includes TAM, architecture guidance, and fastest response times.

Business Support+

👉 “Mid-tier support”

Includes architecture support but no TAM.